

Applied Mathematical Methods for Computing

Module 1

Algebra, vectors and matrices

Module 2

Sequence and Series

Module 3

Combinatorics

Lesson 3.1 Counting

-  Introduction
Video • 1 min
-  Counting
Video • 5 min
-  Complex counting
Video • 4 min
-  Counting
Practice Assignment • Grade: 100%
-  Inclusion–Exclusion Principle
Dialogue • 30 min

3.2 The Pigeonhole Principle

-  The Pigeonhole Principle
Video • 3 min
-  The Pigeonhole Principle – examples part 1
Video • 4 min
-  The Pigeonhole Principle – examples part 2
Video • 3 min
-  Use the Pigeonhole Principle
Practice Assignment • Grade: 100%
-  Think of an example
Discussion Prompt • 20 min
-  **Exercises with hints and tips**
Reading • 1h

Lesson 3.3 Order Matters

-  Permutations
Video • 6 min
-  Permutations - solving a case study of your choice
Practice Assignment • 30 min
-  Permutations
Practice Assignment • 20 min
-  Combinations
Video • 10 min
-  No order/combinations
Practice Assignment • 25 min
-  Combinatorics
Practice Assignment • 40 min

3.4 Summary and Assessments

-  Exercises with hints and tips
Reading • 10 min
-  Conclusion
Video • 3 min
-  Check your understanding: End of module 3
Graded Assignment • 20 min

Module 4

Probability and Statistics

Lesson 4.1 Probability Definition and Properties

-  Introduction to probability
Video • 11 min
-  Probability
Practice Assignment • 30 min

Exercises with hints and tips

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Now it's time to put into practice the concepts you have learnt so far in this module. Attempt the following exercises. If you get stuck, you can refer to the hints and tips at the bottom of the page.

Please note that these exercises are optional, further practice – but we strongly recommend that you engage with them in order to test your knowledge and see where you might need to do additional study.

1. A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.
 - a. How many balls must she select to be sure of having at least three balls of the same colour?
 - b. How many balls must she select to be sure of having at least three blue balls?
2. Show that if seven distinct integers are selected from the first 10 positive integers, there must be at least two pairs of these integers with the sum of 11.
3. Show that if there are 30 students in a class, then at least two have last names that begin with the same letter.
4. Suppose that there are nine students in a class at a small college.
 - a. Show that the class must have at least five male students or at least five female students.
 - b. Show that the class must have at least three male students or at least seven female students.

Hints and tips for each question:

1. Use the extended version of the Pigeonhole Principle. Each colour is a place and the balls are pigeons.
2. Each pair whose sum is 11 is a place and those 10 numbers are pigeons.
3. We have 26 places (each letter is a place) and we have 30 pigeons.
4. There are two places: males and females. Students are our pigeons.

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 **Completed**

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